Fiscal Year:	FY 2005	**Task Last Updated:**	FY 01/28/2005
PI Name:	Kuzmich, Alex Ph.D.		
Project Title:	Entangled atomic clock		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Laser cooled atomic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	30332-0100	**Congressional District:**	5
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2002 Physical Sciences 02-OBPR-03
Start Date:	04/01/2004	**End Date:**	09/30/2006
No. of Post Docs:	2	**No. of PhD Degrees:**	
No. of PhD Candidates:	3	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Kennedy, T.A.Brian (School of Physics)		
Grant/Contract No.:	GSN 4516		
Performance Goal No.:			
Performance Goal Text:			
	A four-year program of experimental and theoretical research on development of entanglement-enhanced atomic clock will be undertaken. Atomic entanglement will be generated via collective QND measurements of populations of the two levels involved in the atomic clock transition. In the first phase of the research, experimental and theoretical investigations will be done with a cold sample of atoms typical of present-day atomic clocks. In the second stage, we will investigate BEC-based entangled atomic clock. For the latter case we will pursue theoretical and experimental investigations of the issues of stability of spatial modes and the impact of the collisional frequency shift on the entanglement-enhanced performance of such a clock. We intend to experimentally implement a quantum non-demolition interaction between our atomic ensemble and a light field. Measurement of the light field amounts to a quantum non-demolition measurement of the atoms, creating an		

Task Description:	entangled state. The essential new idea we propose is the use of a single off-resonant light field which when correctly tuned with respect to the hyperfine levels measures the population difference of the levels levels $|1\rangle $ and $|2\rangle $ associated with the clock transition. Moreover, under these conditions the interaction conserves the population difference of the atomic sample, a vital element in our experimental measurement strategy. Another objective of the research program is to investigate methods which allow cancellation of classical noise sources, such as those associated with distribution of coupling strengths seen by an atomic ensemble in a spatially varying field mode. If uncorrected, such noise could entirely mask entanglement, precluding the possible enhancement in accuracy of an atomic clock. We propose the use of ``matched" observables, which enable large amounts of entanglement to be extracted under these circumstances of apparently weak entanglement. We will investigate the concept of matched observables in the contexts of both cold non-degenerate and BEC-based atomic clocks. The proposed work will significantly impact possible future microgravity atomic clock experiments. Those experiments will be attempting to reach higher accuracy than ground-based atomic clocks. One avenue to even higher accuracy is to increase the number of atoms and/or interaction time. However, this avenue may have already been pushed close to its limits. The development of an entanglement-enhanced atomic clock would represent a major breakthrough and pave the way for further improvements. Measurement-induced entanglement on the atomic clock transition may very well become an integral part of the space-based atomic clock of the future.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	We have realized experimentally the coherent quantum state transfer from a two-level atomic system to a single photon. The entanglement between a single photon (signal) and a two-component ensemble of cold rubidium atoms was used to project the quantum memory element (the atomic ensemble) onto any desired qubit state by measuring the signal field in a suitable basis. The atomic qubit was read out by stimulating the directional emission of a single photon (idler) from the (entangled) collective state of the ensemble. Faithful atomic memory preparation and readout were verified by the observed correlations between the signal and the idler photons. These results enable implementation of distributed quantum networking. The entanglement of multiatom quantum states was considered theoretically. In order to cancel classical noise due to the inhomogeneous light-atom coupling, the concept of matched multiatom observables was proposed. The general approach was illustrated on the example of rotation angle measurement using quantum non-demolition interactions, and it was shown that the multiatom states that were thought to be weakly entangled on account of the classical inhomogeneity noise, can exhibit near maximum entanglement. This approach to eliminate decoherence should be of broad relevance to quantum information processing with atomic ensembles.
Bibliography Type:	Description: (Last Updated: 02/13/2008)
Articles in Peer-reviewed Journals	Kuzmich A, Kennedy TA. "Non-symmetric entanglement of atomic ensembles." Phys Rev Lett. 2004 Jan 23;92(3):030407. Epub 2004 Jan 23. PMID: 14753857 , Jan-2004
Articles in Peer-reviewed Journals	Matsukevich DN, Kuzmich A. "Quantum state transfer between matter and light." Science. 2004 Oct 22;306(5696):663-6. PMID: 15499014 , Oct-2004